

SEBASTIAN CORREA-GALLEGO

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Research Interests

Microbial communities assembled from the same species pool sometimes converge on a single reproducible state and sometimes do not. I want to know what determines which, and whether that can be predicted from how cells divide their finite resources between growth and metabolic capacity. My approach is experimental, combining cultivation, growth physiology, and quantitative analysis of community structure, and my longer-term interest is what assembly can tell us about the evolution of biological individuality.

Education

Universidad EAFIT, Medellín, Colombia

2022–2026

B.Sc. in Biology, conferred June 2026

GPA: 4.44 / 5.00

Relevant coursework: Ecology; Evolution; Microbiology; Biostatistics and Experimental Design; Dynamic Systems; Fundamentals of Computational Biology; Data Science in Life Sciences; Experimental Biology.

Research Experience

Visiting Student Intern

Aug 2025–Jan 2026

[The Evolution of Cells, Symbionts and Organelles Lab](#), Purdue University

Advisor: Sergio A. Muñoz-Gómez, Ph.D.

Direct supervision: Shahed U. A. Shazib, Ph.D.

- Investigated how two osmotrophic yeasts (*K. marxianus* and *V. polyspora*) reorganize proteome investment under carbon limitation, within the project *Proteome Allocation Rules in Osmotrophic Eukaryotes*.
- Generated growth curves across defined carbon regimes for both organisms in triplicate, and estimated cell volume from microscopy images across approximately 2,100 manually measured cells; all analysis in R.
- Contributed to the proteomics workflow from sample preparation to integrative analysis, and calibrated an automated continuous-culture system (chemostat and turbidostat modes) for the lab's steady-state work.

Undergraduate Thesis Researcher

2024–2026

Universidad EAFIT

Advisor: Nicolás Pinel Peláez, Ph.D.

Thesis: [Cultivable Microbial Community Structure Along a Light Gradient in a Tropical Volcaniclastic Cave](#)

- Tested whether light availability, as a proxy for energy input, structures the cultivable microbial community of the Organal San Antonio, a tropical volcaniclastic cave at ~2350 m a.s.l. in Támesis, Colombia, establishing a spatially resolved baseline for an organal-type pseudokarstic system.
- Designed and executed three field campaigns, advancing from exploratory sampling to a gradient design with three light-defined sectors and five replicates each, and sought external funding through two competitive schemes.
- Found an approximately sixfold reduction in cultivable density from the entrance to the aphotic sector, with composition partitioning by zone (PERMANOVA $R^2 = 0.57$) and roughly a quarter of the transition assemblage growing nowhere else; comparison across sectors relied on a morphotype crosswalk whose resolution limit is stated explicitly in the thesis.

Manuscripts

Proteome Allocation Rules in Osmotrophic Eukaryotes

Coauthored manuscript in preparation

The Evolution of Cells, Symbionts and Organelles Lab, Purdue University

Conferences and Presentations

Cultivable Microbial Community Structure Along a Light Gradient in a Tropical Volcaniclastic Cave

Oral Presentation

May 2026

V Simposio de Biología Divulgativa, Universidad EAFIT, Medellín, Colombia

Proteome Allocation Rules in Osmotrophic Eukaryotes

Oral Presentation

Jan 2026

UREP-C and UREP-LA Final Symposium, Purdue University, West Lafayette, IN, USA

Characterization of Cave-Dwelling Phototrophic Microbial Communities from the Northwestern Andes

Poster Presentation

Nov 2024

II Simposio de Biología Divulgativa, Universidad EAFIT, Medellín, Colombia

Positioning Microorganisms as Key Actors in the Maintenance of Andean Forests through Science Communication

Poster Presentation

May 2024

XXIII Encuentro Departamental de Semilleros de Investigación, RedCOLSI, Medellín, Colombia

Exploration of Organel Ecosystems in the Organel San Antonio (Támesis, Antioquia) and Identification of the Psychrophilic Photosynthetic Microorganisms Inhabiting Them

Poster Presentation

May 2023

XXII Encuentro Departamental de Semilleros de Investigación, RedCOLSI, Medellín, Colombia

Exploration of the Thermoacidophilic Microorganisms Inhabiting the Aguas Calientes Hot Spring

Poster Presentation

Sep 2022

Feria de Semilleros de Investigación, Universidad EAFIT, Medellín, Colombia

Fellowships and Grants

Simons Graduate Fellowships in Ecology and Evolution

Under review

Simons Foundation

Application submitted June 2026; notification expected October 2026.

Research Internship Sponsorship

2025

Fundación Fraternidad Medellín

Full funding for a six-month research internship at Purdue University (UREP-C Program).

Undergraduate Scholarship

2022–2026

Fundación Fraternidad Medellín

Full tuition scholarship for the B.Sc. in Biology, awarded on academic merit.

Honors and Awards

Stanford Biology Preview Program

Sep 2026

Department of Biology, Stanford University

Selected participant, 2026 cohort.

Second Prize, II Concurso de Relatos Científicos

May 2025

Biology Program, Universidad EAFIT

Western Cordillera category, for the narrative essay *Ethos de una Palma Andina*.

Second Prize, I Concurso de Relatos Científicos

Nov 2024

Biology Program, Universidad EAFIT

Eastern Cordillera category, for the narrative essay *La Clínica de la Sombra*.

Second Prize, Poster Presentation

Nov 2024

II Simposio de Biología Divulgativa, Universidad EAFIT

Academic Service

Student Coordinator, Research Group on Microbiology and Astrobiology (SIAB) *Jan 2024–Jun 2025*
Universidad EAFIT, affiliated with the Research Group on Geosciences and Biodiversity (GEBI)
Member since January 2022, initially through the University’s social-service requirement. Led a student research group on microbiology, astrobiology, and extremophiles, working with members to learn and apply microbiological techniques and explore the diversity of these organisms.

Teaching, Mentoring and Outreach

Volunteer Science Instructor and Materials Developer *2025–2026*
Institución Educativa de Desarrollo Rural Miguel Valencia, Jardín, Colombia

- Completed two 30-hour social-service projects (60 hours total) in support of the lower-secondary Natural Sciences course at a rural public school, in a mixed in-person and remote format.
- Accompanied the classroom teacher during instruction and developed digital teaching materials on the origin of the universe, microbiology, evolution, and biological diversity.

Project Monitor, Science-Outreach Initiative on Microorganisms and Andean Forests *2024*
Research Group on Microbiology and Astrobiology (SIAB), Universidad EAFIT

- Assumed day-to-day leadership of the initiative during its final six months, working with a designer to produce science-communication material on the role of microorganisms in the maintenance of Andean forests for the El Globo nature reserve (Támesis, Colombia).

Logistics Volunteer, Universo Semilleros Research Fair *Aug 2024*
Universidad EAFIT

- Provided logistical support at an institutional research-presentation event organized by the Vice-Rectorate for Science, Technology and Innovation.

Certifications

Responsible Conduct of Research (RCR) Training *Aug 2025*
CITI Program, Credential ID 71597371, valid through Aug 2029

Agricultural Production Technician *2019–2021*
Servicio Nacional de Aprendizaje (SENA), Colombia

Technical Skills

Laboratory and field methods: microbial cultivation and isolation, continuous-culture operation (chemostat and turbidostat), growth-curve measurement, proteomics sample preparation, light and epifluorescence microscopy (MitoTracker and DAPI staining) with image-based cell measurement, environmental sampling, and ecological field records. Coursework-based bench training in molecular biology and biochemistry.

Quantitative and computational tools: R (multivariate statistics, ecological data analysis, visualization), QGIS, and $\LaTeX$; Python and Linux/bash (coursework-based).

Languages: Spanish (native); English (professional working proficiency).